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THE PALAEOMAGNETIC STABILITY AND THE PETROLOGY OF SOME CAENOZOIC AND CRETACEOUS ANDESITES OF BULGARIA *

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Some different young Bulgarian andesites were tested by the palaeomagnetic stability tests. It was found that e.g. andesites of the same age but from different localities possess a strong difference of stability. Some reasons for this phenomenon are cleared by application of other magnetic and non-magnetic methods (e.g. thermomagnetic and ore-microscopical ones).

1. INTRODUCTION

During 1963, 1965 and 1967 palaeomagnetic expeditions have been carried out by the authors. These were the first palaeomagnetic investigations of Bulgarian rocks. The purpose of the investigations was to compare the results of these measurements with similar ones obtained elsewhere, particularly in the German Democratic Republic, and also to give some contributions to palaeomagnetic and stratigraphic correlations of geological complexes of Bulgaria. The first results are published by Nozharov [3], particularly for basalts.

In this paper we shall present only results of the stability tests of NRM for andesites. To determine the palaeomagnetic stability we used complex methods, some of which were non-magnetic ones, the results of which will be discussed (see also Vollstädt [4-6]).

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2. GEOLOGICAL DESCRIPTION OF ROCKS

The collection of investigated samples includes 5 groups of rocks:

1. Latito-andesites from the region of Jabalkovo, Haskovsko. The samples B-68 to B-73 were collected in an old quarry in the surroundings of Kairjaka. The effusion in this region presents two lava flows, one upon the other. The upper lava flow which was investigated, is bright-grey small-porphyry latito-andesite, and the lower one is reddish, perhaps secondly altered latito-andesite. To the west of the old quarry there is a new one, the rocks in which are coarse-grained pyroxene-biotite latito-andesites. According to Ivanov [1], the investigated latito-andesites from Jabalkovo could belong, respectively, to the 'horizon of the first middle-acid volcanism Ol_1^1', or to the 'first latito-andesite effusion' at the end of Priabonian and the beginning of Oligocene.

2. Andesito-basalts from the region of Gledka, Kardshali. The samples B-60 to B-67 are from a quarry near the highway between Kardshali and Momchilgrad. The andesito-basalts are dark blue, even black rocks,

Table 1
Intensities, directions and pole positions of the Bulgarian andesites.

Site no.	Location	Rock type	Age	Mean vector R	D_m	I_m	k	α_{95}	Polarity	φ'	λ'	δ_p	δ_m
B-43 to B-47	Madzharevo 41°N; 25°E	Trachy-andesite	Oligocene	4.485	10	+51	8	29	+	76	160°E	0.5	39
B-60 to B-67	Gledka 41°N; 25°E	Andesito-basalt	Oligocene	7.952	202	−57	148	5	−	88	173°E	0.1	6
B-68 to B-73	Jabalkovo 42°N; 27°E	Latito-andesite	Oligocene	5.946	214	−74	93	7	−	62	172°E	0.2	12
B-74 to B-79	Varlibrjag 42°N; 27°E	Andesite	Senon	5.812	12	38	26	13	M	66	177°E	9	15

containing a great amount of volcanic glass. According to Ivanov these rocks could belong, respectively, to the 'horizon of the third middle-acid volcanism Ol_2^3 - b', or to the 'andesito-basalt effusion' from the cyclical volcanism in the East-Rhodope palaeogenic subsidance. Consequently, these rocks are upper Oligocene in age.

3. Andesites from the region of mine Varlibrjag, Burgasko. The samples B-74 to B-79 are collected from a natural exposure in the region near the mine. These andesites are grey-green rocks. According to geologists these rocks are Cretaceous (Senonian).

4. Andesites from the region of Madzharevo, Haskovsko. The samples B-30 to B-35 were collected in a road-cut at the highway about 500–800 m from Madzharevo.

5. Trachy-andesites from the region of Madzharevo, Haskovsko. The samples B-43 to B-47 were collected along the highway Madzharevo – Ivailovgrad, about 1.5–2 km from the village of geologoprospectors. According to Ivanov the andesites and trachy-andesites from Madzharevo could belong, resp., to the 'horizon of the second middle-acid volcanism Ol_2^1 - b', or to the 'middle latito-andesite effusion' in the Oligocene.

Unfortunately, we cannot give a more detailed geological and petrographical characteristic of the above mentioned rocks, because of lacking of such data in the geological literature.

3. RESULTS OF PALAEOMAGNETIC INVESTIGATIONS

Natural remanent magnetization (NMR) of samples was measured using an astatic magnetometer (according to Blackett) and a Dolginov magnetometer. The measurements of NRM and the stability tests were carried out independently in the Geomagnetic Institute, Potsdam, and in the Geophysical Institute, Sofia.

Table 1 shows the average intensities and directions of NRM and the pole positions for several andesite types. It is especially noticed that the intensities for the latito-andesite Jabalkovo, and andesito-basalt Gledka are higher compared to the others.

The character of stability was determined applying three known methods and having in mind some remarks by Kropacek [2]: AC partial demagnetization and demagnetization by DC magnetic fields and by temperature.

The thermal and AC partial demagnetization of the latito-andesite Jabalkovo (B-68 to B-73) are shown in fig. 1. Both curves show very stable behaviour. At temperatures above 400°C, and at fields above 400 Oe the sample still possesses 90% of the magnetization and 70% of the original values, respectively. The investigation in DC magnetic field gives $H_c' = 100$ Oe.

The samples B-60 to B-67 (andesito-basalt, Gledka) have similar properties. The demagnetization curves of these samples show in each case 50% of their NRM after 400°C and 400 Oe, as shown in fig. 2. H_c' for these samples is 30 Oe.

The AC partial demagnetization curve (fig. 3) for andesite Varlibrjag (B-74 to B-79) shows that it has metastable behaviour. At fields above 200 Oe the inversely magnetized sample loses its viscous magnetization and only 50% of the original magnetization remained. On the other hand, thermal demagnetization did not give useful results; the curve is very compli-

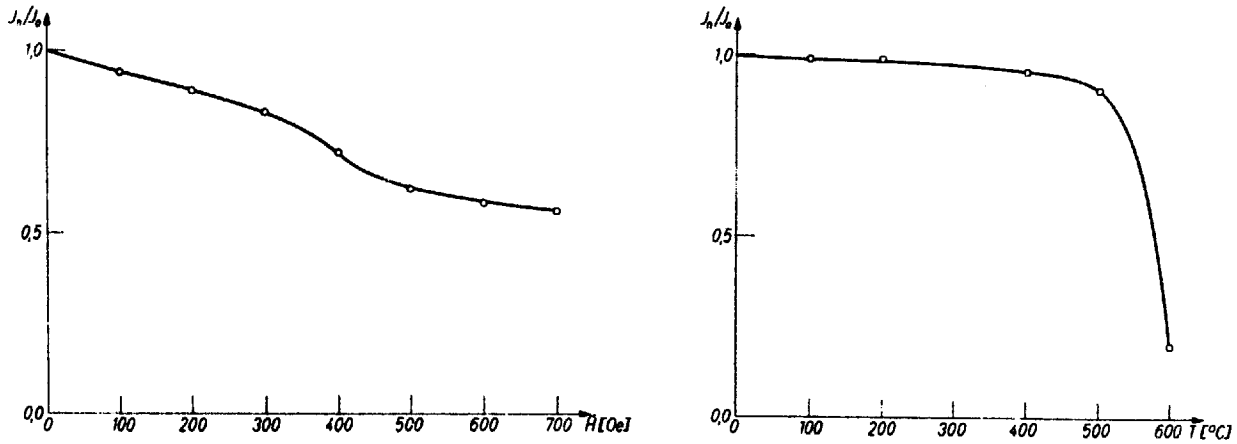


Fig. 1. AC (a) and thermal (b) demagnetization of NRM (andesite, Jabalkovo).

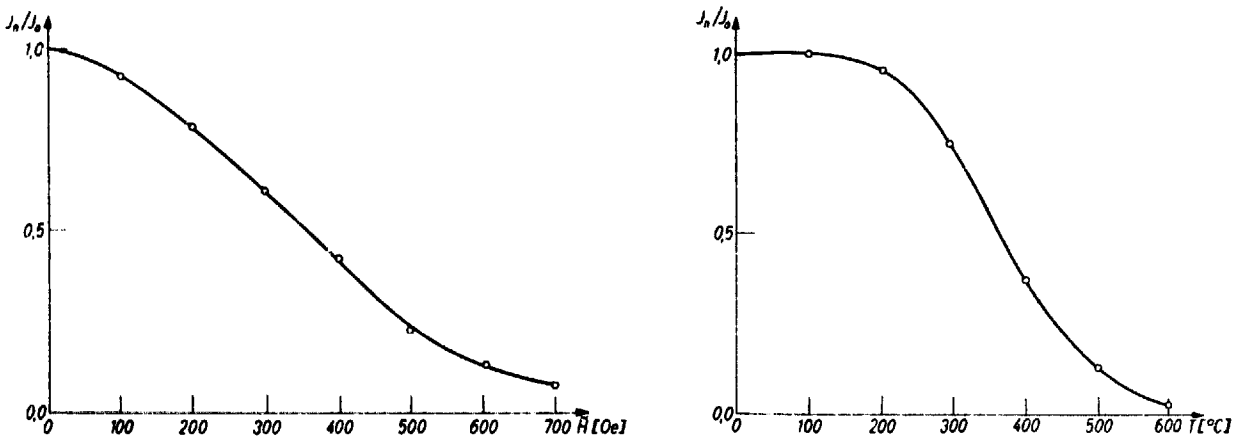


Fig. 2. AC (a) and thermal (b) demagnetization of NRM (andesito-basalt, Gledka).

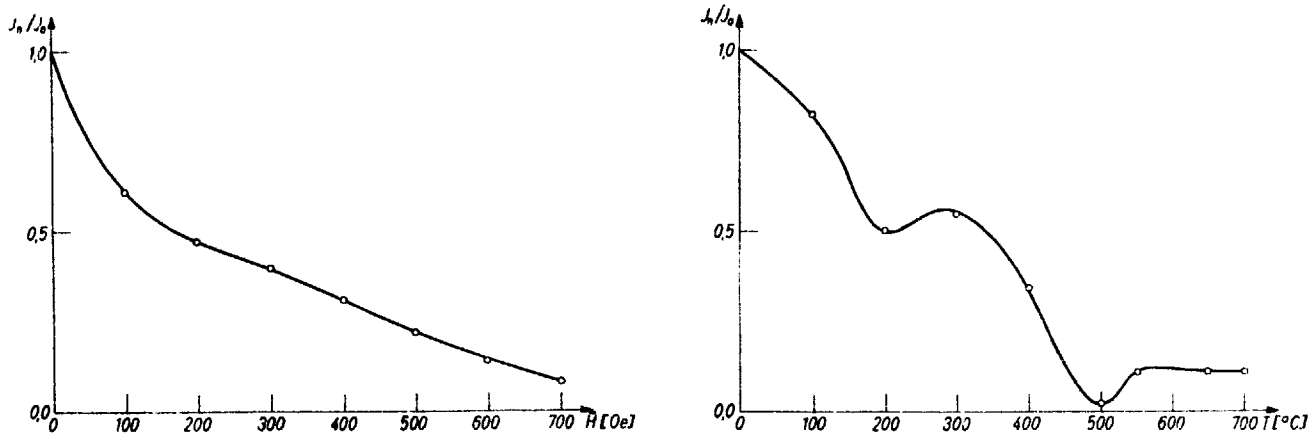


Fig. 3. AC (a) and thermal (b) magnetization of NRM (andesite, Varlibrjag).

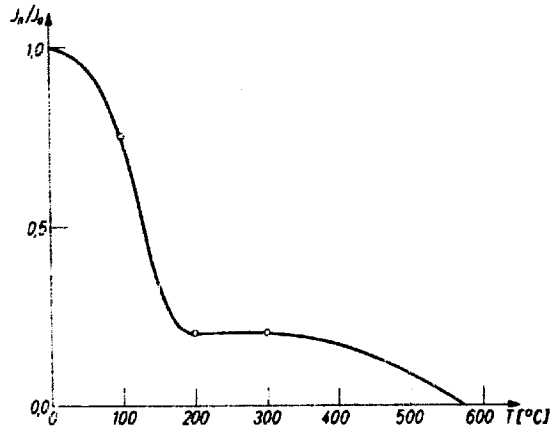


Fig. 4. Thermal magnetization of NRM (andesite, Madzharevo).

cated (see section 4). H'_c for these samples is about 11–12 Oe.

The stability tests of the remaining samples of andesite (B-30 to B-35) and trachy-andesite (B-43 to B-47) from Madzharevo show that they are unstable. The results are shown in fig. 4. The curve decomposes into two parts in the case of andesite Madzharevo, because of viscous magnetization. At temperatures above 400°C only 10% of the original NRM remains. Demagnetization by direct magnetic fields gives H'_c = 5 Oe. Later on we shall discuss this further.

It is clear that all demagnetization methods must be used for a thorough investigation of the sample for palaeomagnetic purposes, in order to be able to characterize the stability of magnetization and to determine its origin. However, results obtained by demagnetization by an alternating magnetic field have the greatest importance.

Now we can classify the investigated samples into three types according to their stability:

latito-andesite, Jabalkovo	– stable
andesito-basalt, Gledka	– stable
andesite, Varlibrjag	– metastable
trachy-andesite, Madzharevo	– unstable to metastable
andesite, Madzharevo	– unstable

These properties of stability express themselves in keeping the direction of NRM. Figs. 5a-e show declination D and inclination I for several andesites. Both stable samples are inversely magnetized. After magnetic cleaning at 200°C and 200 Oe the direction of the samples makes a small circle of confidence contrary to the recent geomagnetic field (see table 1). This similar nature is shown by samples B-43 to B-47 and B-74 to B-79, although the deviation from average is already greater. Moreover, the trachy-andesite (B-43 to B-47) is normally magnetized. The values of direction for the sample andesite Madzharevo are very scattered and it is impossible to determine a representative result.

4. DISCUSSION

The 'classical' magnetic tests of stability, described above, can clarify only partially the reasons of stability or unstability. It was necessary to use other methods to determine the differences in the stability between the andesites. Such complex investigations were carried out by Vollstädt [4-6] using caenozoic basalts and phonolites. In this paper only some of the possible methods and their results are described. These methods are: investigation by ore microscope, thermomagnetic balance and X-ray diffractometer.

4.1. Andesito-basalt, Gledka

We examined polished surfaces and found homogeneous titano-magnetite (we used oil immersion at a magnification of about 500). The grains are very small (about 10 μ) and in most cases idiomorphic. The grains are homogeneously distributed in the matrix. In spite of using very high magnification we did not find oxidations and intergrowths. The thermomagnetic curve (see fig. 6, B-62) confirms this result. The presence of only one Curie point at 560°C and the relative high saturation magnetization refer to a titano-magnetite with a small amount of TiO_2. X-ray measurements show that the titanomagnetite is similar to pure magnetite, since the D values and the lattice constant a_0 of the separate samples are slightly different from the magnetite one. The results are shown in table 2. According to Vollstädt [6] all these properties represent criteria for a stable behaviour. The high stability in respect to temperature influences is striking; it shows a small difference between the heating and cooling curves of saturation magnetization.

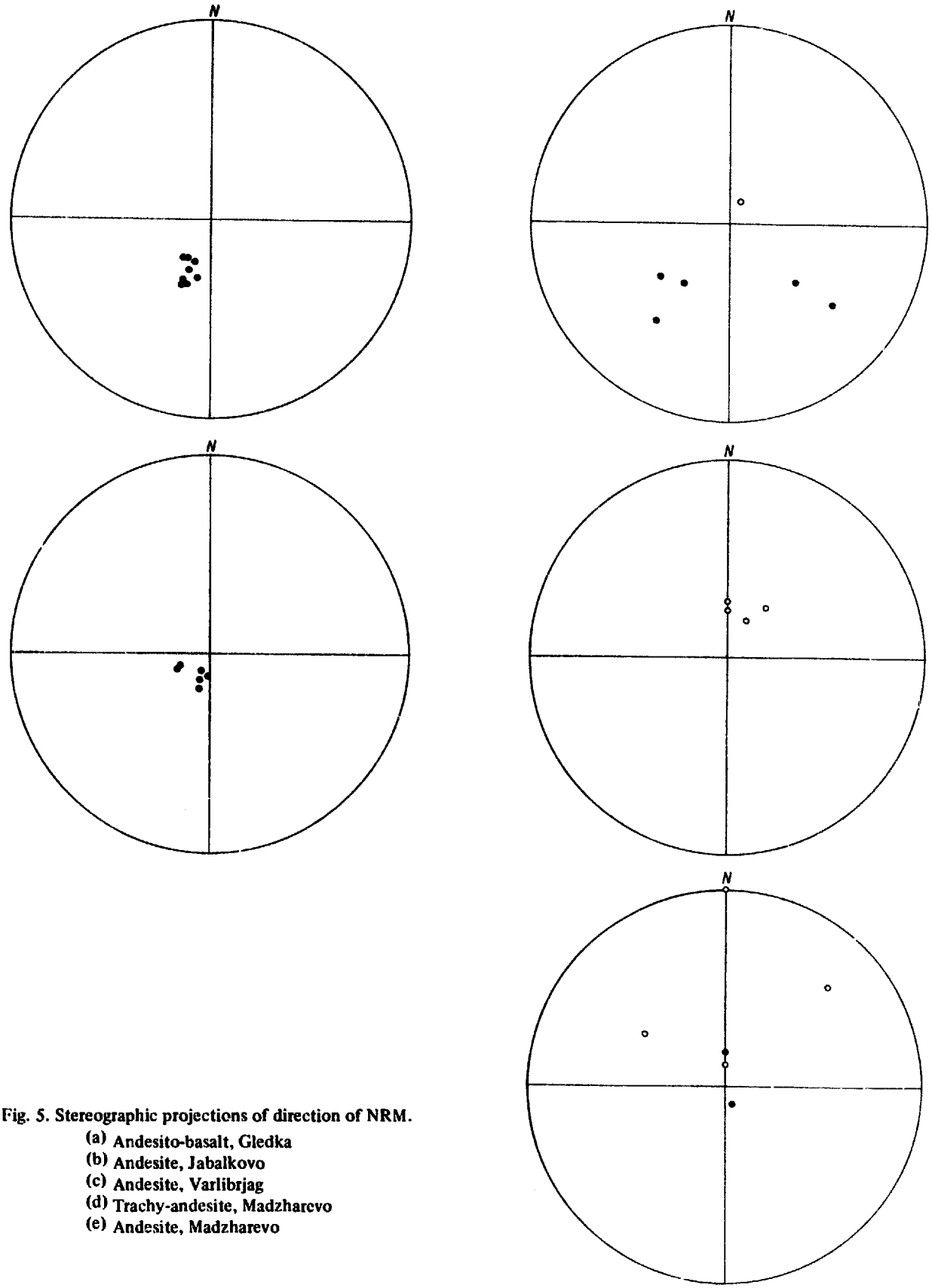


Fig. 5. Stereographic projections of direction of NRM.
(a) Andesito-basalt, Gledka
(b) Andesite, Jabalkovo
(c) Andesite, Varlibrjag
(d) Trachy-andesite, Madzharevo
(e) Andesite, Madzharevo

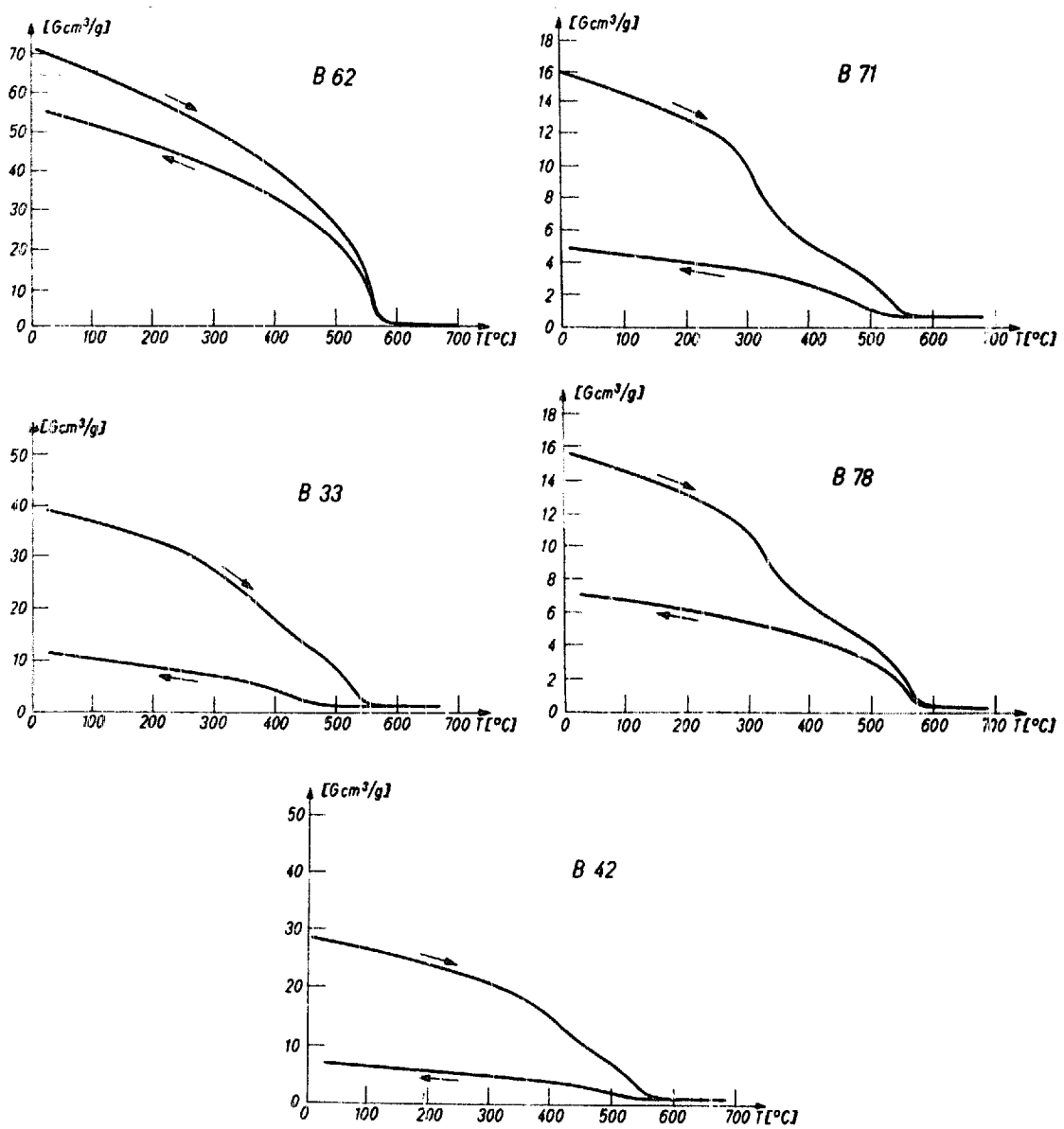


Fig. 6. Thermomagnetic curves of some Bulgarian andesites.

4.2. **Latito-andesite, Jabalkovo**

Under the microscope this sample shows a very interesting feature: only small remnants appear of a primary titanomagnetite. It is altered into ilmenite with magnetite interstices, and ilmenite in turn changes into titanohematite. An investigated grain is shown in fig. 7. The X-ray diagram of this magnetically separated sample showed minerals from the complex hematite-ilmenite (see table 3). This corresponds with results obtained by ore-microscopic investigations. The amount of titanomagnetite was not determined since its portion was too small. The intense intergrowths of this hemo-ilmenite with the small titanomagnetite grains permitted us to separate the paramagnetic (antiferromagnetic) material. Contrary to the X-ray results, the thermomagnetic curve

Table 2
X-ray results of the samples B-62 and B-78.

B-62		B-78		Magnetite
D	*I*	*D*	*I*	D_{ma}
		3.717	1	
2.958	4	2.992	2.5	2.97
		2.720	2.5	
2.519	10	2.543	10	2.53
2.086	5	2.105	7	2.097
		1.846	2	
1.712	2	1.701	2	1.714
1.613	4	1.617	2.5	1.615
1.483	8	1.485	4	1.484
1.323	1			1.326
1.278	2	1.270	1	1.278
1.212	1.5			1.209
1.120	1			1.120
1.092	7			1.091
1.050	4			1.048

Table 3
Results of X-ray investigations of sample B-71 (latito-andesite, Jabalkovo). D_{me} = *D* (measured), D_{he} = *D* (hematite), D_{il} = *D* (ilmenite).

D_{me}	*I*	D_{he}	D_{il}
3.697	2.5	3.68	3.70
3.009	1		
2.717	10	2.69	2.74
2.525	7.5	2.51	2.53
2.210	3.5	2.20	2.23
1.842	2.5	1.837	1.865
1.696	4	1.691	1.720
1.603	1	1.596	1.630
1.486	4	1.484	1.504
1.453	3	1.451	1.465
1.310	1.5	1.309	1.347

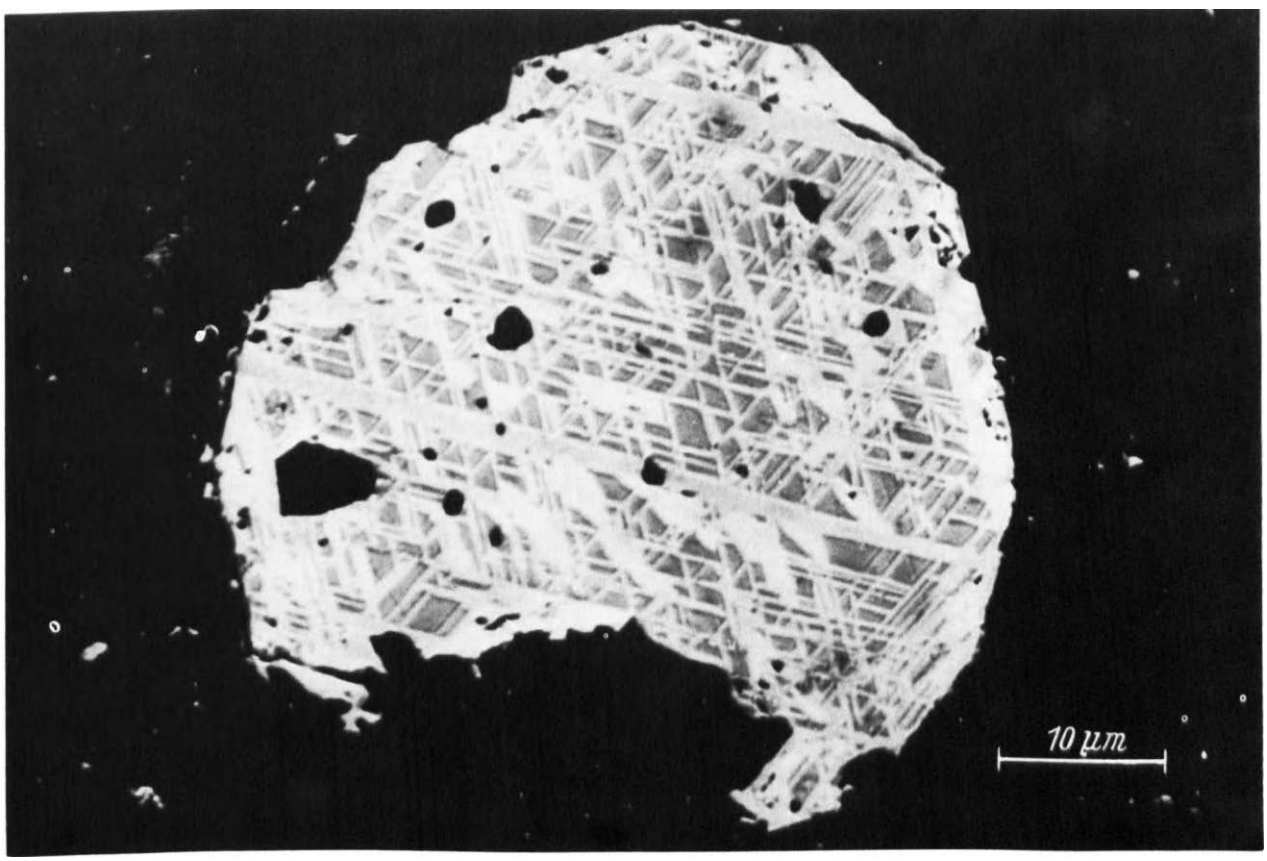


Fig. 7. Mineral of the ilmenite-hematite series with exsolutions (sample B-71).

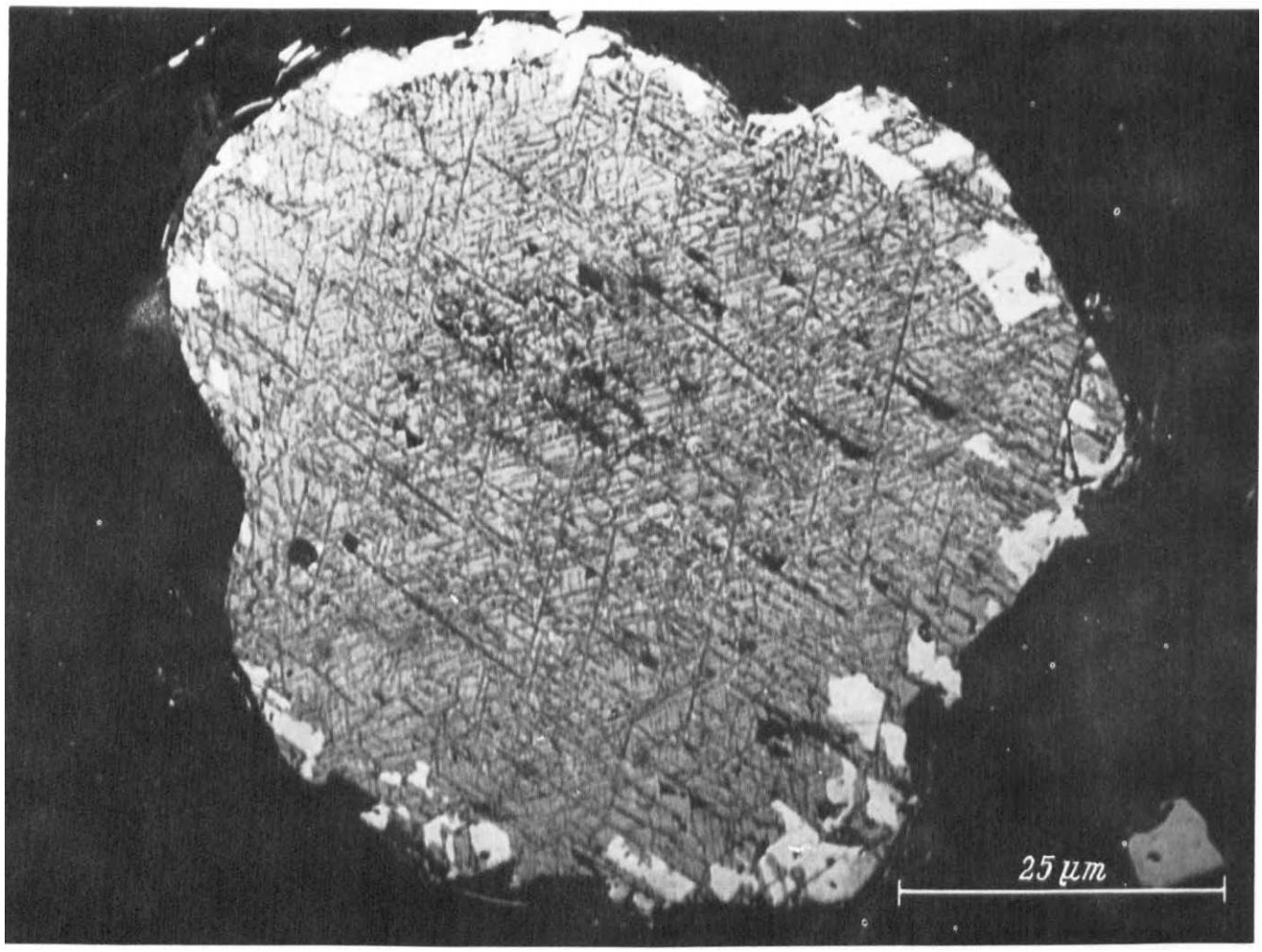


Fig. 8. Titanomagnetite with ilmenite-lamellae along (111) and spinel needles along (100) and small hematite (sample B-78).

shows that there is titanomagnetite (see fig. 6, B-71). It has two peaks which represent the Curie points of hemoilmenite and titanomagnetite. The cooling curve has only one Curie point, which is due to the higher hematite amount produced by the heating to 700°C in air. The stable properties of this sample are due to the predominant amount of hematite and ilmenite. The results obtained on these samples confirm the observations which one of the authors [6] made on tertiary basalts and the serials of the works of Watkins and Haggerty. Watkins and Haggerty [7] found direct connections between primary oxidation state and magnetic properties. The oxidation state of our complex corresponds to the oxidation index *V* according to Watkins and Haggerty.

4.3. Andesite, Varlibrjag

In this sample we found relative great titanomagnetite grains (about 100 μ), which are oxidized in most cases. There are two systems of lamellae in the titanomagnetite: one, ilmenite, along (111) planes, and the second one consists of fine 'needles' of spinel exsolved along the (100) cube faces. This is the first stage of high-temperature oxidation (see also Wilson and Haggerty [8]). Beside this, the grains are martitized in the outermost rim (see fig. 8). The big grains, their xenomorphic form and the high-temperature oxidations can be the reason for the metastable character.

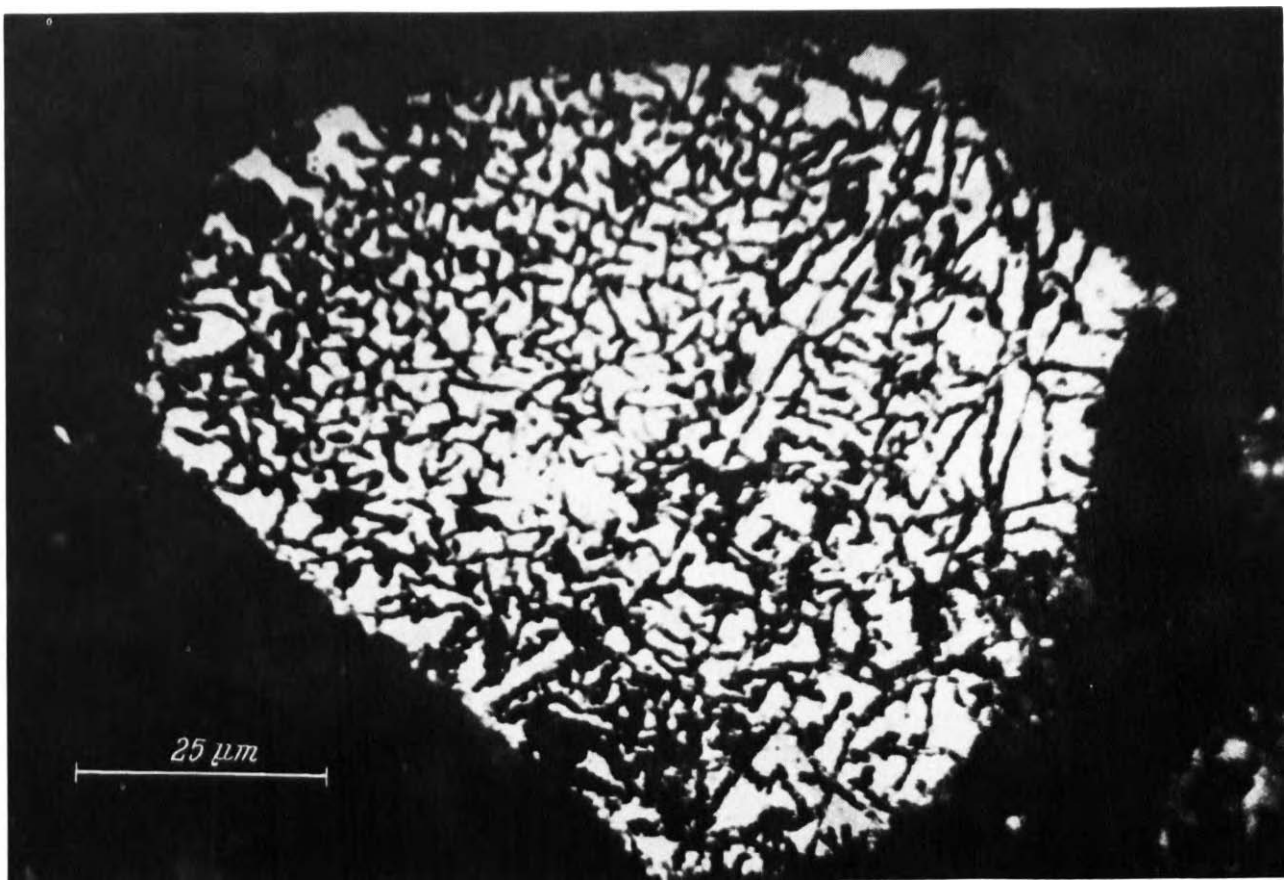


Fig. 9. Irregular resolution phenomena in titanomagnetite (sample B-33).

4.4. Trachy-andesite and andesite, Madzharevo

Both unstable samples have similar composition of magnetic particles. They represent homogeneous titanomagnetite, which has very xenomorphic forms and big grains. The titanomagnetite grains of the most unstable andesite, Madzharevo, show odd irregular solution phenomena, contributing to the instability (fig. 9, B-33). The saturation magnetization is very small, of course. Both X-ray diagrams point out only titanomagnetite. The Curie point is 540°C, which suggests a small amount of TiO_2.

If, finally, we consider the character of the direction of NRM we get the following picture: three samples are inversely magnetized, namely latito-andesite Jabalkovo, andesite Varlibrjag and andesito-basalt Gledka. The sample trachy-andesite Madzharevo shows normal direction of NRM. This fact agrees with results of statistical measurements produced by Wilson et al. [8]. They have found that the most normal magnetized samples contain homogeneous or slightly altered magnetic minerals. On the other hand, the minerals of inverse samples show mostly a stage of high-temperature oxidation. In our measurements only the andesite-basalt Gledka shows properties contrary to the results of Wilson et al.: although it is inversely magnetized it contains homogeneous titanomagnetite and no oxidation phenomena.

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